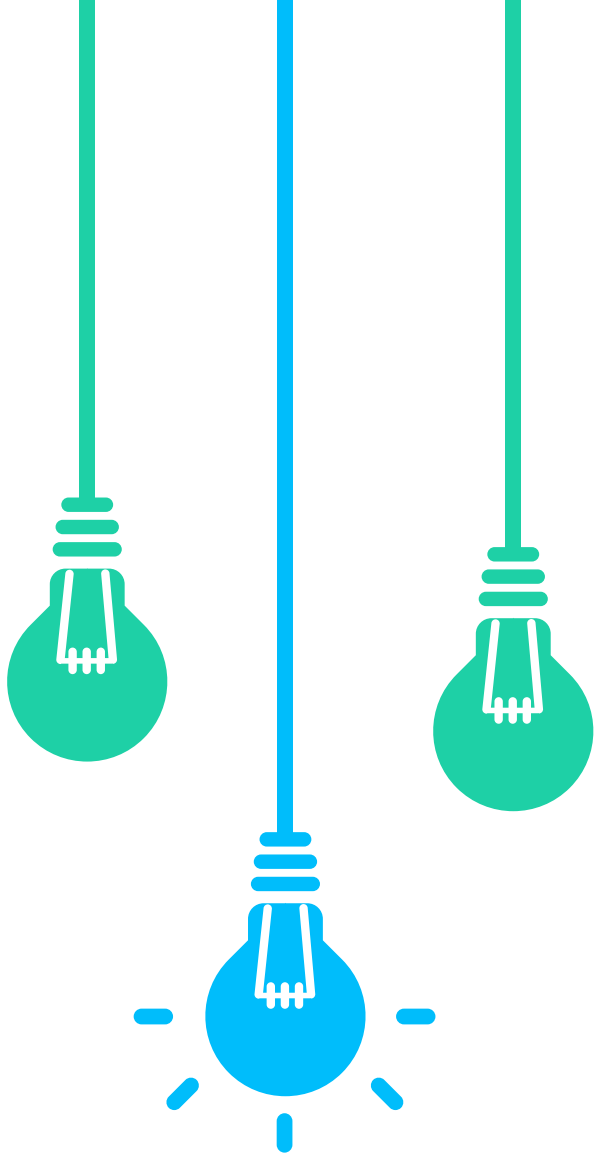
Security and Privacy of IOT Devices in Smart HealthCare



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Motivation

1. The integration of medical “things” - devices, smart sensors, mHealth apps, artificial intelligence (AI) etc. had increased the usage of IoT devices
2. This has urged the demand for various medical wearable devices globally as people can easily access and monitor their health at any given time
3. The hospitals segment held a global internet of things (IoT) in healthcare market share of about 34.6% in 2020
4. Medical devices pose a unique cybersecurity risk in that attacks or hacks can directly endanger patient privacy and safety
5. Identify the open-research challenges in security and privacy of Smart HealthCare System

Introduction

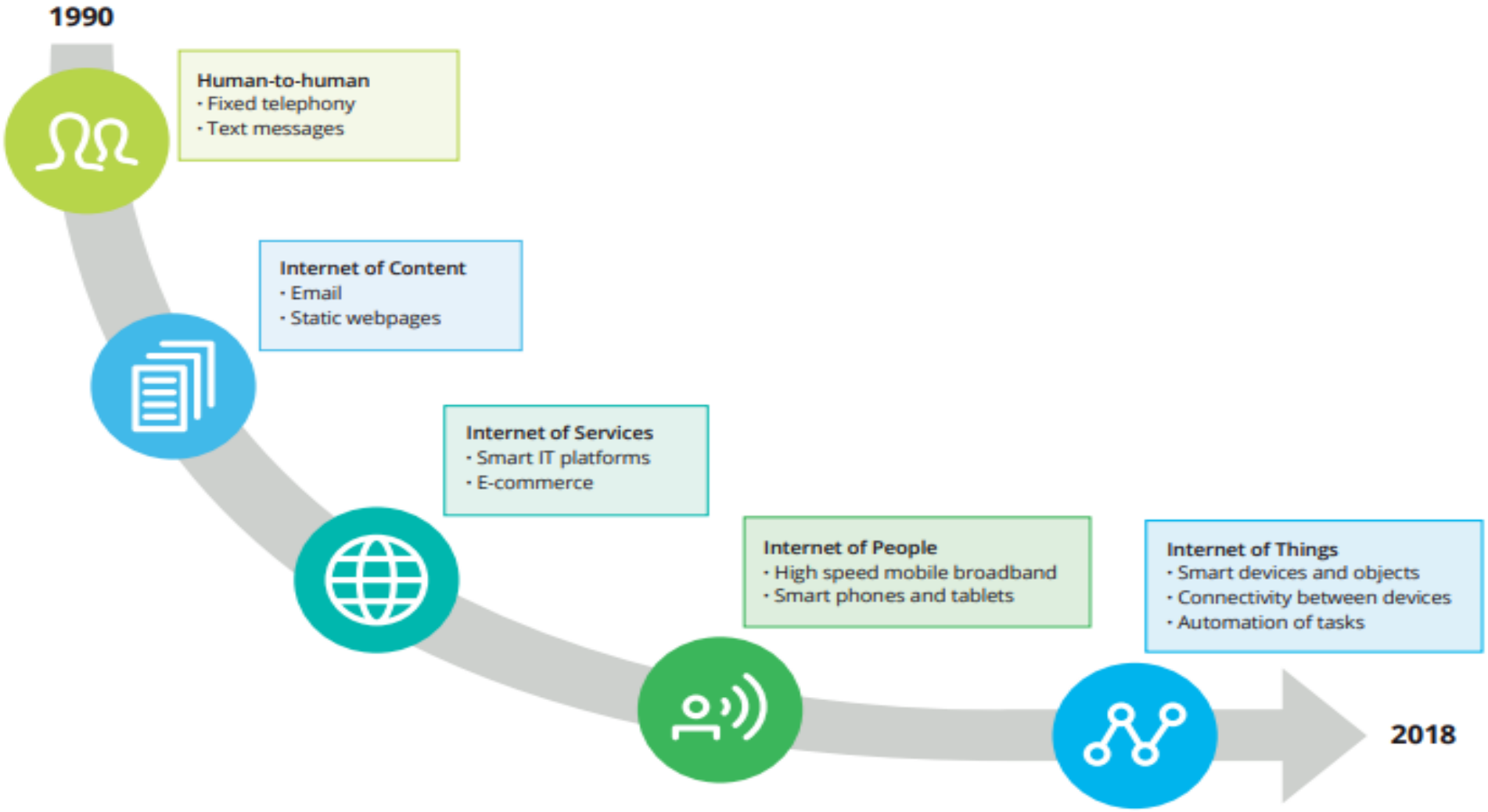


Fig. 1. Technological and cultural changes enabling the development of the Internet of Things

Introduction (Cont.)

- The Internet of Things (IoT) consists of Subsets,
- The Internet of Healthcare Things (IoHT) is the convergence and integration of sensor data collected by medical devices and mobile technologies and applied in Healthcare
- Devices linked to cloud platforms on which captured data is stored and analyzed has come to be known as the Internet of Medical Things (IoMT)
- Most of the connected medical devices can only send information or can send and receive data.
- The US Food and Drug Administration (FDA) was made aware of 11 cybersecurity vulnerabilities that, if exploited by a remote attacker, could put critical medical devices and networks at risk.

Literature Survey

- In the [3], the author discusses the importance of Iot in HealthCare and how the devices are subjected to Cyber attacks.
- DDos attacks in IOT devices
- Integration of Various Devices
- Medjacking – Hacking of Medical Devices.
  - In June 2015 – TrapX revealed Medjacking
  - In the first case, a blood gas analyzer infected with two different types of malware was used to steal passwords for other hospital systems and confidential data
  - At another hospital, the radiology department's image storage system was used to gain access to the main network, sensitive data was retrieved and sent to a location in China.
  - Stolen medical identities are much more valuable than the price of a stolen credit card number.

Literature Survey

- The Data Encryption technology - Cryptography is a security technology for information exchange and communication. [4]
- Privacy Disclosure of the original data
- Data Search
  - Searchable Symmetric Encryption (SSE)
  - public-key encryption with keyword search (PEKS)

Table 4: Original data.

Height	Weight	Age	Sensitive data
172	63	27	Time serials
178	75	34	Time serials
180	72	26	Time serials
185	77	22	Time serials

Table 5: Anonymity result of original data.

Height	Weight	Age	Sensitive data
17*	**	**	Time serials
17*	**	**	Time serials
18*	7*	2*	Time serials
18*	7*	2*	Time serials

* and ** represent anonymous information.

Privacy Disclosure [4]

Literature Survey

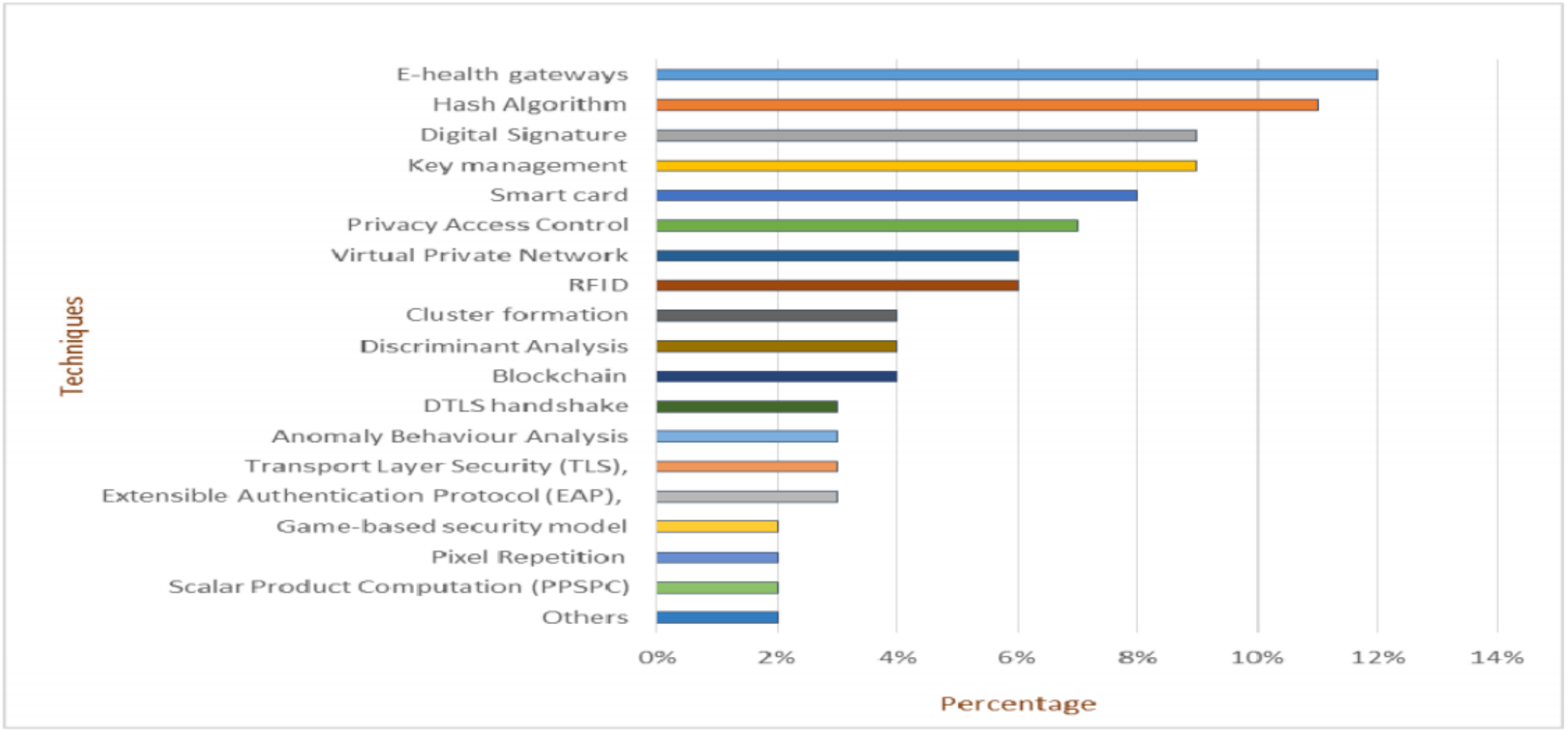


Fig. 2. Classification by security techniques [1]

Current Status

- In Pune, India, June 07, 2021 (GLOBE NEWSWIRE) - The global internet of things (IoT) in healthcare market size is expected to experience significant growth by reaching $ 446.52 billion by 2028 while exhibiting the Compound Annual Growth Rate (CAGR) of 25.9% between 2021 and 2028.
- Cisco Blog - On an average a hospital has 15 to 20 IoMT devices which could be in the form of a smart bed, an IV pump, an MRI machine, a ventilator, a monitor and other health tech devices.

Current Status

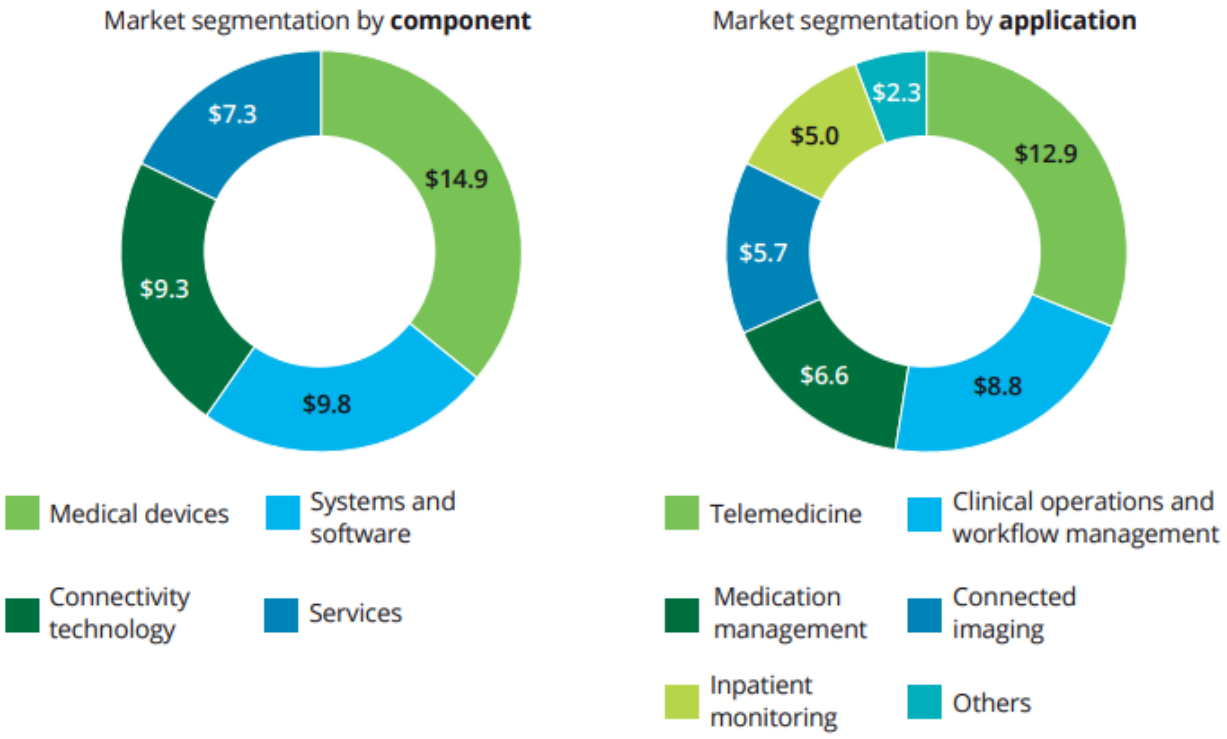


Fig. 3. IoMT market segmentation

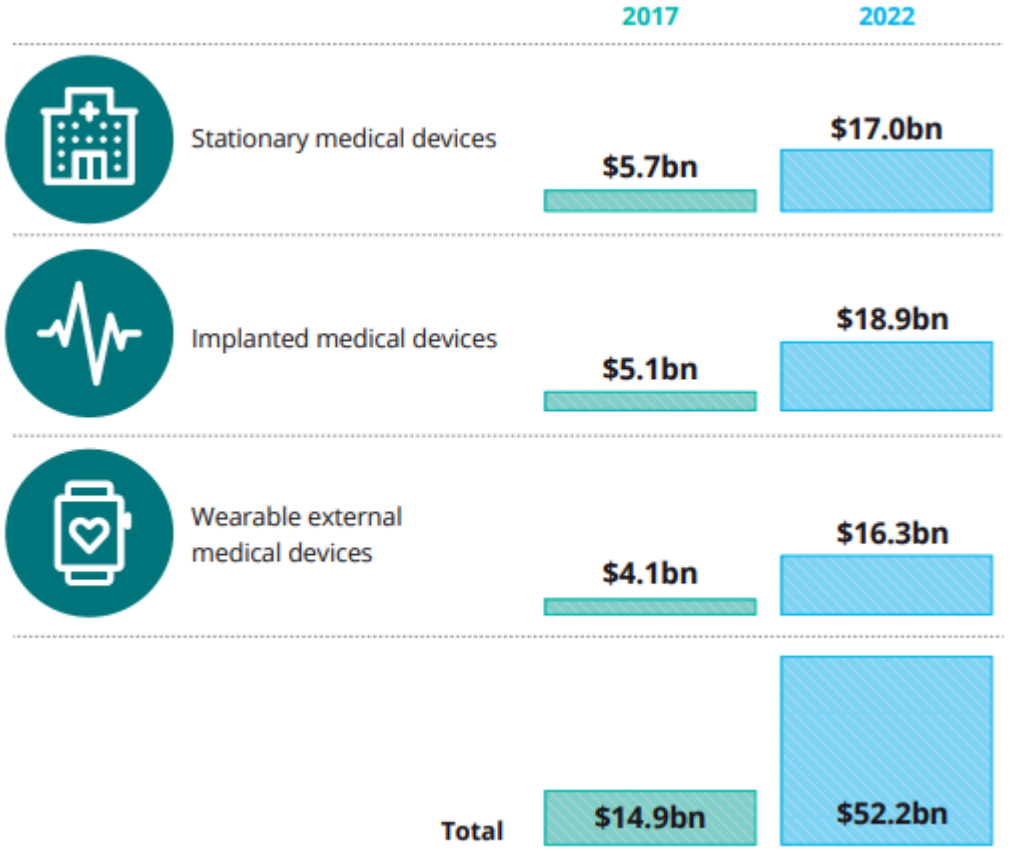


Fig. 4.The market for connected medical devices

Source: Deloitte

Current Status

- The Role of Firmware in Attacking IoMT Devices – A WRECK bug firmware that gives the cyber criminals an easy pass to control the medical devices even without the internet

Few steps to protect internet-connected equipment s from Cyberattacks are

- IT teams should be knowledgeable about IoMT devices
- Strong Passwords
- Network Segmentation
- Active Monitoring of Network Traffic

-- TECH TIMES

Implementation

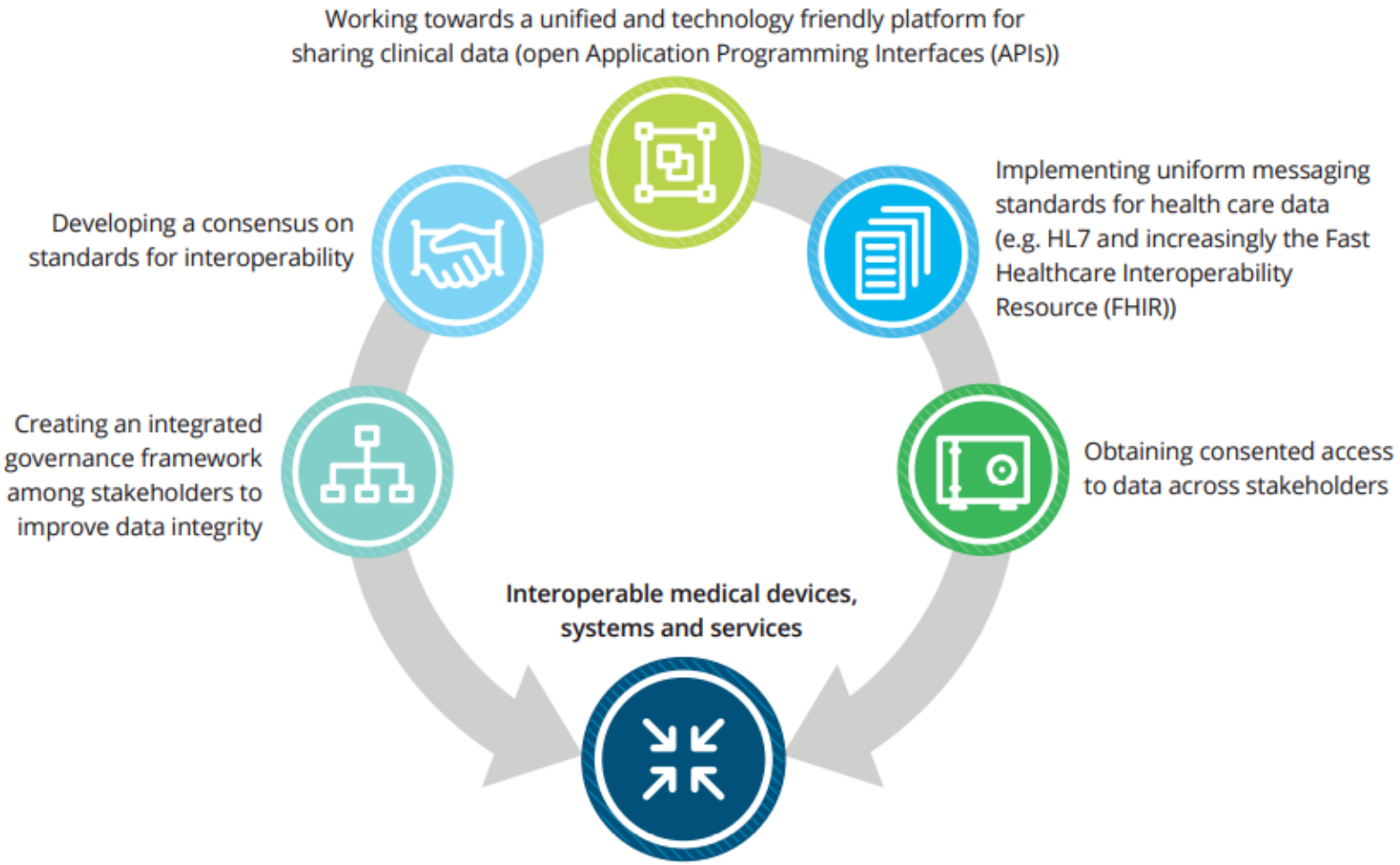


Fig. 5. Key actions life sciences and health care stakeholders when tackling the interoperability challenge

Source: Deloitte LLP, 2018

Implementation

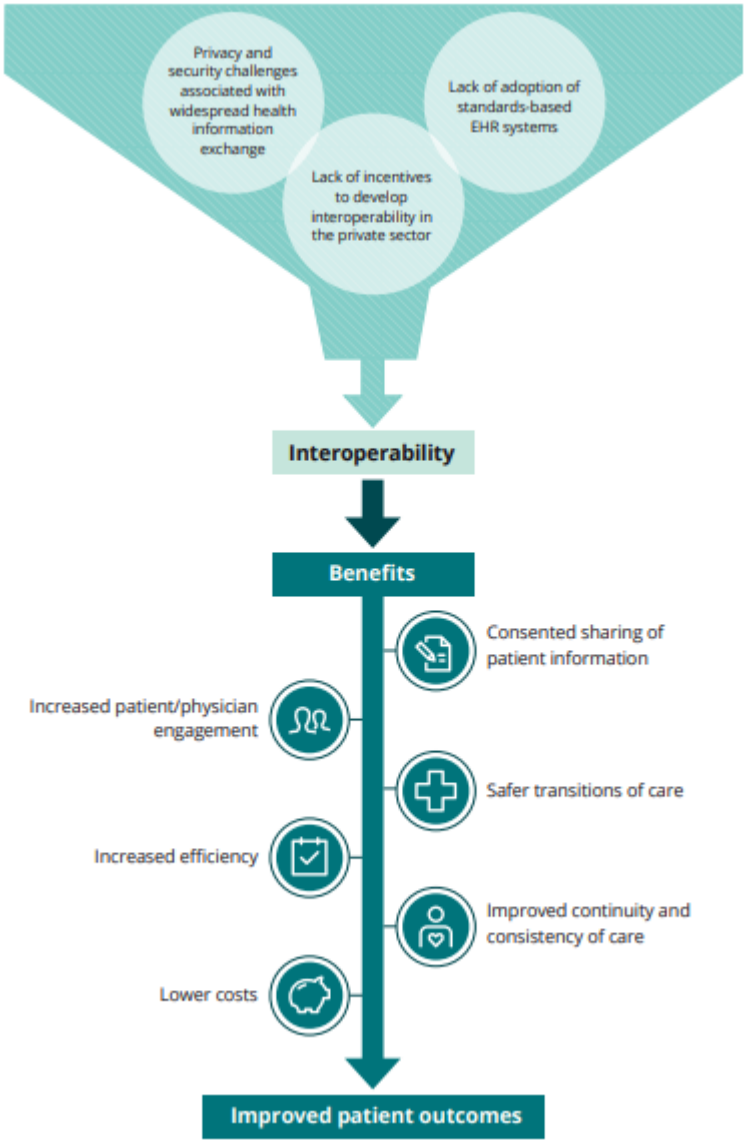


Fig. 6. Barriers to and benefits of interoperability

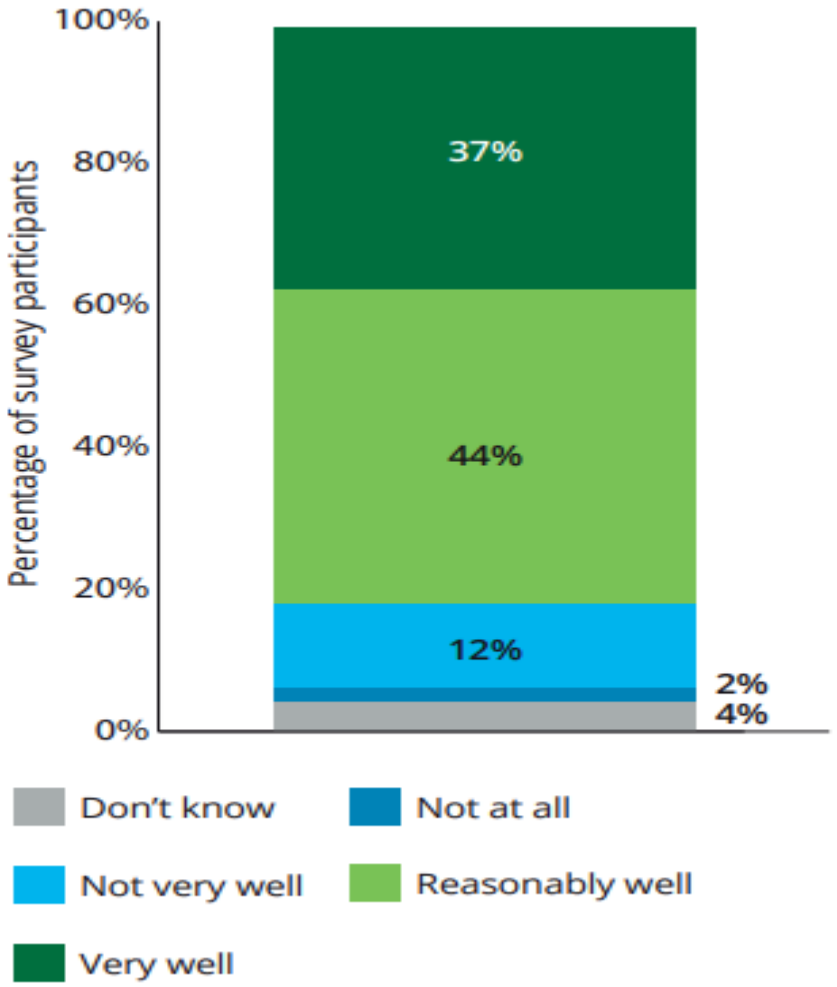


Fig. 7. Preparedness of medtech companies maintaining the cybersecurity of connected medical devices

Implementation

IBM introduces Cognitive IOT devices as a part of Security in Medical devices by introducing wearables

- **Instrumented and Interconnected -** Integrating structured data with unstructured data (e.g. social media, text, email, voice)
- **Intelligent -** Applying adaptable machine learning techniques to interconnected physical device data patterns and Trends
- **In Real-time -** Responding dynamically to chronic, multi-dimensional and/or time sensitive problems
- **In Context Data -** Creating a personal and consumable experience by understanding the human element in the context of both structured and unstructured data

Source: IBM MD&I

Implementation



Improve Operations, Efficiency & Convenience

- Reduce the need for multiple devices through hands-free operations
- Re-engineer existing business processes, improving productivity
- Increase efficiency, and reduce operational errors and costs



Increase Safety

- Monitor and coach on the wellness and safety of the user
- Prevent and detect hazards, and notify employees and supervisors



Increase Quality

- Provide rich data points for operational and customer understanding
- Measure final product size and shape, and verify specifications
- Predict defects to optimize machines
- Create new personalized customer service interactions



Improve Collaboration & Training

- Provide remote training and real-time feedback
- Enable faster decision-making and responsiveness

Source: IBM MD&I

Conclusion

- Making these Wearables interoperable can enhance the existing methodologies
- The wearable devices through their hospital or prescribed from the doctor are recommended to minimize Cyber attacks
- Usage of BYOD should be minimized

Future Work

- Providing data security and privacy at all stages of data is essential
- IT teams should be knowledgeable about IoMT devices
- Active Monitoring of Network Traffic
- Device Patching

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Thank You